

Step-by-Step Guide to 1NF, 2NF, 3NF, and BCNF

Introduction

Normalization is a process in database design used to reduce data redundancy and improve data integrity. It involves organizing attributes and relations in a way that minimizes anomalies.

We check normal forms in the following order:

$$1NF \rightarrow 2NF \rightarrow 3NF \rightarrow BCNF$$

A relation must satisfy lower normal forms before qualifying for higher ones.

1 First Normal Form (1NF)

Definition

A relation is in 1NF if:

- All attributes have atomic (indivisible) values.
- There are no repeating groups or arrays.

Steps to Check

1. Identify attributes with multi-valued or composite data.
2. Ensure each attribute contains only atomic values.

Example (Valid)

$R(\text{StudentID}, \text{Name}, \text{Course})$

Data:

(101, Alice, Math), (102, Bob, Physics)

All attributes are atomic — relation is in 1NF.

Example (Invalid)

(101, Alice, {Math, Physics})

‘Course’ has multiple values in one cell — not in 1NF.

2 Second Normal Form (2NF)

Definition

A relation is in 2NF if:

- It is in 1NF.
- No **partial dependency** exists — no non-prime attribute depends on part of a candidate key.

Steps to Check

1. Find candidate keys.
2. Identify prime attributes (part of a candidate key).
3. Check if any non-prime attribute is dependent on a subset of a candidate key.

Example (Invalid)

$R(\text{StudentID}, \text{CourseID}, \text{StudentName})$

Candidate key: $\{\text{StudentID}, \text{CourseID}\}$ FD: $\text{StudentID} \rightarrow \text{StudentName}$ (partial dependency) Not in 2NF.

Example (Valid)

Split into:

$R_1(\text{StudentID}, \text{StudentName}), \quad R_2(\text{StudentID}, \text{CourseID})$

Now, no partial dependency — in 2NF.

3 Third Normal Form (3NF)

Definition

A relation is in 3NF if:

- It is in 2NF.
- No **transitive dependency** exists — no non-prime attribute depends on another non-prime attribute.

Steps to Check

1. Check for 2NF.
2. Identify if any non-prime attribute depends on another non-prime attribute.

Example (Invalid)

$$R(\text{StudentID}, \text{DeptID}, \text{DeptName})$$

FDs:

$$\text{StudentID} \rightarrow \text{DeptID}, \quad \text{DeptID} \rightarrow \text{DeptName}$$

Here, $\text{StudentID} \rightarrow \text{DeptName}$ via DeptID — transitive dependency.

Example (Valid)

Split into:

$$R_1(\text{DeptID}, \text{DeptName}), \quad R_2(\text{StudentID}, \text{DeptID})$$

Now in 3NF.

4 Boyce-Codd Normal Form (BCNF)

Definition

A relation is in BCNF if:

- It is in 3NF.
- For every functional dependency $X \rightarrow Y$, X is a superkey.

Steps to Check

1. Verify 3NF.
2. For each FD $X \rightarrow Y$, check if X is a superkey.

Example (Invalid)

$$R(\text{Teacher}, \text{Course}, \text{Room})$$

FD: $\text{Course} \rightarrow \text{Room}$ If ‘Course’ is not a superkey, BCNF is violated.

Example (Valid)

Split into:

$$R_1(\text{Course}, \text{Room}), \quad R_2(\text{Teacher}, \text{Course})$$

Now all FDs have superkey determinants — in BCNF.

Summary Table

Normal Form	Key Condition
1NF	Atomic values, no repeating groups
2NF	1NF + No partial dependency
3NF	2NF + No transitive dependency
BCNF	3NF + Every determinant is a superkey